Department of Computer Science and Engineering (Data Science)
B.Tech. Sem: III Subject: Statistics for Data Science
Experiment 1

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Date:	**Experiment Title: Data Structures in Python**
Aim	To study data types in Python and its functions.
Software	Google Colab
Implementation	Question: Write a comment in Python. Code:

```
In [5]: #Welcome
```

Question: Write a multiline comment/paragraph in Python.
Code:

```
In [6]: '''print("Hello")
        print("Welcome to the program")
        '''
        print("Hi there")

Hi there
```

Primitive Data Types
Question: Write a program to print an integer, float, string, complex number, Boolean, and bytes in Python and display their data type.
Code:

```
In [9]: z1=4
        z2=4.2
        z3=True
        z4='Python'
        a=complex(3,4)
        print(z1,z2,z3,z4,a)
        print(type(z1),type(z2),type(z3),type(z4),type(a))

4 4.2 True Python (3+4j)
<class 'int'> <class 'float'> <class 'bool'> <class 'str'> <class 'complex'
```

Data Structures

Lists

Question: Write a program to create a list. Collect heterogenous data in it.

Code:

```
In [1]: l=[13,"Program",True]
        print(l)

[13, 'Program', True]
```

Question: Write a program to print a list.

Code:

```
In [1]: l=[13,"Program",True]
        print(l)

[13, 'Program', True]
```

Question: Write a program to print a new list. Append an item in this list.

Code:

```
: l=[13,"Program",True]
  print(l)
  l.append("Python")
  print(l)

[13, 'Program', True]
[13, 'Program', True, 'Python']
```

Question: Write a program to make a copy of the previous list.

Code:

```
In [14]: l=[13,"Program",True]
         b=l.copy()
         print(b)

[13, 'Program', True, 'Python program']
```

Question: Write a program to concatenate 2 lists and print the output.
Code:

```
In [3]: l1=[6,"Name",False]
        l2=["Place",15]
        l1.append(l2)
        print(l1)

[6, 'Name', False, ['Place', 15]]
```

Question: Write a program to count the number of elements present in a list.

```
In [30]: print(len(l1))

3
```

Code:

Question: Write a program to print the length of a list.

```
In [30]: print(len(l1))

3
```

Code:

Question: Write a program to append more than 1 item in a list.
Code:

```
In [4]: l2.extend(['Python','C'])
        print(l2)

['Place', 15, 'Python', 'C']
```

Question: Write a program to extend a list.
Code:

```
In [4]: l2.extend(['Python','C'])
        print(l2)

['Place', 15, 'Python', 'C']
```

Question: Write a program to insert a value at a position in a list.
Code:

```
In [37]: l1=[6,"Name",False]
         l1.insert(2,"Hi")
         print(l1)

[6, 'Name', 'Hi', False]
```

Question: Write a program to delete a value at a given position in a list.
Code:

```
In [30]: l1=[6,"Name",False,"Hi"]
         l1.pop(3)
         print(l1)

[6, 'Name', False]
```

Question: Write a program to remove a value from the list.
Code:

```
In [40]: l1=[6,"Name",False]
         l1.remove(6)
         print(l1)

['Name', False]
```

Question: Write a program to slice the data in a list.

Code:

```
In [12]: l1=["Hi","I","am","Bhuvi",54,False,18,"Statistics","Experiment"]
         print(l1[3:4])

['Bhuvi']
```

Question: Write a program to slice data in a list using positions.
Code:

```
In [12]: l1=["Hi","I","am","Bhuvi",54,False,18,"Statistics","Experiment"]
         print(l1[3:4])

['Bhuvi']
```

Question: Write a program to print the last 8 elements.
Code:

```
In [41]: l1=["Hi","I","am","Bhuvi",54,False,18,"Statistics","Experiment"]
         print(l1[-1:-8])

[]
```

Question: Write a program to print the last value of a list.
Code:

```
In [33]: l1=["Hi","I","am","Bhuvi",54,False,18,"Statistics","Experiment"]
         print(l1[-8:-1])

['I', 'am', 'Bhuvi', 54, False, 18, 'Statistics']
```

Question: Write a program to print the central value of a list.
Code:

```
In [35]: l1=["Hi","I","am","Bhuvi",54,False,18,"Statistics","Experiment"]
         middle=(len(l1)-1)/2
         print(l1[int((len(l1)-1)/2)])

54
```

Tuples

Question: Write a program to create a tuple. Collect heterogenous data in it.
Code:

```
In [44]: l=(13,"Program","Tuple")
         print(l)
         print(l[0])

(13, 'Program', 'Tuple')
13
```

Question: Write a program to print the position of an item in the tuple.
Code:

```
In [46]: l2=(13,"Program","Tuple")
         print(l2)
         print(l2.index("Program"))

(13, 'Program', 'Tuple')
1
```

Question: Print a new tuple. Write a program to concatenate two tuples.
Code:

```
In [58]: l1=(13,"Program","Tuple")
         l2=("Hi","Good Morning","How are you?")
         l3=l1+l2
         print(l3)

(13, 'Program', 'Tuple', 'Hi', 'Good Morning', 'How are you?')
```

Question: Write a program to print the value at position 2 in the concatenated tuple.
Code:

```
In [59]: print(l3[2])

Tuple
```

Question: Write a program to change the element of a tuple.

Code:

```
In [62]: l=list(l2)
         print(l)
         l[2]="Wake up"
         print(tuple(l))

['Hi', 'Good Morning', 'How are you?']
('Hi', 'Good Morning', 'Wake up')
```

Dictionary

Question: Write a program to create and print a dictionary.

Code:

```
In [66]: k={"Maharashtra":"India","New York":"USA"}
         print(k)

{'Maharashtra': 'India', 'New York': 'USA'}
```

Question: Write a program to print values of a dictionary using keys.

Code:

```
k={"Maharashtra":"India","New York":"USA"}
print(k["Maharashtra"])

India
```

Question: Write a program to create a multidimensional dictionary.

Code:

```
In [32]: k={"Maharashtra":{"City":"Mumbai","Weather":"Cold"},"New York":"USA"}
         print(k)

{'Maharashtra': {'City': 'Mumbai', 'Weather': 'Cold'}, 'New York': 'USA'}
```

Question: Write a program to print values from the multidimensional dictionary using keys.

Code:

```
In [70]: k={"Maharashtra":{"City":"Mumbai","Weather":"Cold"},"New York":"USA"}
         print(k["Maharashtra"]["City"])

Mumbai
```

If-Else Condition

Question: Brother is 12 years old. Sister is 15 years old. Write a program that prints who is older using if-else statement.

Code:

```
In [74]: brother=12
         sister=15
         if(sister>brother):
             print("Sister is elder")
         else:
             print("Brother is elder")

         Sister is elder
```

Question: Take the input of ages from the user. Write a program that prints who is older using if-else statement.

Code:

```
In [3]: z1=input("Enter age of a")
        z2=input("Enter age of b")
        if(z1>z2):
            print("z1 is elder")
        else:
            print("z2 is older")

        Enter age of a14
        Enter age of b15
        z2 is older
```

For Loop

Question: Write a program that prints the elements of a list using for loop.

Code:

```
In [1]: l1=[6,"Name",False]
        for i in l1:
            print(i)

6
Name
False
```

Question: Write a program that enumerates and prints the elements of a list using for loop.
Code:

```
In [4]: l1=[6,"Name",False]
        for i,j in enumerate(l1):
            print(i,j)

0 6
1 Name
2 False
```

Functions
Question: Write a program to create a function.
Code:

```
In [15]: def function(a):
             if(a%2==0):
                 return(a)
             else:
                 return(0)
         function(5)

Out[15]: 0
```

Question: Create a function that adds two numbers.

Code:

```
In [16]: def function(a,b):
             z1=a+b
             print(z1)
         function(5,3)

         8
```

Question: Create a function that adds two numbers. Take input from the user.
Code:

```
In [19]: def function(a,b):
             z1=a+b
             print(z1)
         z2=int(input("Enter a number"))
         z3=int(input("Enter a number"))
         function(z2,z3)

         Enter a number20
         Enter a number15
         35
```

Question: Create a function that adds two strings. Take input from the user.
Code:

```
In [20]: def function(a,b):
             z1=a+b
             print(z1)
         z2=input("Enter a string")
         z3=input("Enter a string")
         function(z2,z3)

         Enter a stringHi
         Enter a stringHello
         HiHello
```

Sets
Question: Write a program to create and print a set.

Code:

```
In [21]: z={"Hello",1,"Hi",False}
         print(z)

{'Hello', 'Hi', 1, False}
```

Question: Write a program to print a set with duplicate values.
Code:

```
In [25]: z={"Hello",1,"Hi",False,"Hi"}
         print(z)

{False, 'Hi', 'Hello', 1}
```

Question: Write a program to print the length of a set.
Code:

```
In [22]: print(len(z))

4
```

Question: Write a program to create a set and print its data type.
Code:

```
In [23]: z={"Hello",1,"Hi",False}
         print(type(z))

<class 'set'>
```

Question: Write a program to check if a set takes duplicate values with different capitalization/formatting.

	Code:
	In [24]: `z={"Hello",1,"Hi",False,"hi"}`
	`print(z)`
	`{False, 1, 'hi', 'Hello', 'Hi'}`

Conclusion	Hence, we have studied and implemented data structures in Python.

Signature of Faculty